

# Chi-Yao (Trey) Huang

Email: [cy.huang@asu.edu](mailto:cy.huang@asu.edu) | LinkedIn: [linkedin.com/in/chi-yao-huang](https://www.linkedin.com/in/chi-yao-huang)

Website: [huang-chiyao.github.io](https://huang-chiyao.github.io) | GitHub: [github.com/huang-chiyao](https://github.com/huang-chiyao) | Phone: +1 (623) 258-7190

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## SUMMARY

Ph.D. researcher in computer vision, machine learning, and robotics, specializing in representation learning, 3D perception, and foundation models. My research develops interpretable and controllable latent-space frameworks that connect geometry, semantics, and generation toward general-purpose and embodied AI. Former Senior R&D Engineer at HTC VIVE with experience developing VO/SLAM tracking systems for commercial VR/AR products.

## EDUCATION

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| <b>Arizona State University</b>                                | Tempe, AZ         |
| • <i>Ph.D. in Computer Science, Advisor: Prof. Yezhou Yang</i> | 2023 – Present    |
| • <i>M.S. in Robotics and Autonomous Systems</i>               | 2021 – 2023       |
| <b>National Taiwan University</b>                              | Taipei, Taiwan    |
| • <i>M.S. in Mechanical Engineering</i>                        | 2015 – 2017       |
| <b>National Sun Yat-Sen University</b>                         | Kaohsiung, Taiwan |
| • <i>B.S. in Mechanical Engineering</i>                        | 2010 – 2014       |

## SELECTED PUBLICATIONS

- **Domain Expansion: A Latent Space Construction Framework for Multi-Task Learning (ICLR 2026):** Proposed an orthogonal latent-space pooling framework to reduce representation collapse and gradient conflict in multi-task learning. Evaluated on ShapeNet and MPIIGaze, showing improved task separation, interpretability, and compositional generalization. ([project](#) | [paper](#) | [code](#) | [talk](#))
- **VOCAL: Visual Odometry via ContrActive Learning (WACV 2026):** Developed a contrastive learning framework for visual odometry by organizing latent representations according to camera motion. Evaluated on KITTI and demonstrated interpretable motion-aware latent structure for pose estimation. ([project](#) | [paper](#) | [talk](#) | [demo](#))

## EXPERIENCE

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| <b>Arizona State University</b>                                                                                                                                                                                                                                   | Tempe, AZ           |
| • <i>Ph.D. Researcher, Computer Vision and Robotics – Prof. Yezhou Yang</i>                                                                                                                                                                                       | Aug 2021 – Present  |
| – Developing interpretable and controllable latent-space frameworks that connect 3D geometry, semantics, and generation for general-purpose and embodied AI.                                                                                                      |                     |
| – Researching representation learning methods for 3D perception, visual odometry, pose estimation, and foundation-model-based visual representations.                                                                                                             |                     |
| – Designed contrastive learning and orthogonal latent-space methods to improve representation structure, task separation, and motion-aware visual understanding.                                                                                                  |                     |
| – Supported by Toyota Research Institute of North America (TRINA).                                                                                                                                                                                                |                     |
| <b>HTC VIVE, Advanced and Creative Team</b>                                                                                                                                                                                                                       | New Taipei, Taiwan  |
| • <i>Senior R&amp;D Engineer, Team Lambda – VR/AR Tracking and Spatial Computing</i>                                                                                                                                                                              | Sep 2017 – Feb 2021 |
| – Developed VO/SLAM and VIO tracking systems for commercial VR/AR products, including VIVE COSMOS, VIVE FOCUS 3, VIVE XR Elite, and VIVE FLOW.                                                                                                                    |                     |
| – <b>VIVE COSMOS:</b> Architected a multi-camera tracking system for consumer VR, achieving sub-millimeter tracking precision in controlled evaluation; developed a high-performance “mothership” SLAM system. ( <a href="#">product</a>   <a href="#">demo</a> ) |                     |
| – <b>VIVE FOCUS 3:</b> Led prototyping for a flagship MR system and optimized a tightly coupled visual-inertial SLAM pipeline on resource-constrained embedded platforms, achieving a 4× runtime speed-up. ( <a href="#">product</a> )                            |                     |
| – <b>VIVE XR Elite:</b> Engineered low-latency VIO and obstacle avoidance systems for mixed-reality interaction; designed core data structures for real-time spatial computing. ( <a href="#">product</a> )                                                       |                     |
| – <b>VIVE FLOW:</b> Designed an inside-out tracking algorithm for limited-FoV sensors and developed a prototype motion-tracking pipeline integrating real-time hand tracking. ( <a href="#">product</a> )                                                         |                     |
| – <b>Scene:</b> Built voxel-based obstacle mapping and semantic mapping modules for VR safety and real-world interaction.                                                                                                                                         |                     |
| – Collaborated with hardware, firmware, and product teams to deploy perception systems under real-time, CPU, and thermal constraints.                                                                                                                             |                     |
| <b>NTU Robotics Lab, National Taiwan University</b>                                                                                                                                                                                                               | Taipei, Taiwan      |
| • <i>Graduate Researcher and Vice System Manager – Prof. Han-Pang Huang</i>                                                                                                                                                                                       | Sep 2015 – Jun 2017 |

- Conducted research in SLAM, 3D reconstruction, dense mapping, and robot motion planning.
- Managed robotic equipment including robotic arms and mobile robots, and developed orientation materials for lab members.
- **Company of Air Defense Artillery Battalion, R.O.C. Army** Taiwan  
*Battalion Dispatcher* Sep 2014 – Sep 2015
  - Coordinated military vehicle and equipment dispatch during training and field exercises.

## SELECTED PROJECTS

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- **Aerial Pose Estimation – Toyota Research Institute of North America:** Developed a UAV pose estimation framework using IR and IMU sensor fusion for high-altitude drone perception under challenging lighting and weather conditions. ([demo](#))
- **3D Landmark Reconstruction for Robot Localization and Mapping:** Implemented a learning-based 3D reconstruction pipeline that jointly optimizes camera trajectories and object-level semantic poses for robot localization. ([demo](#))
- **3D Reconstruction and Path Planning with Signed Distance Function:** Fused dense image information with sparse feature points for efficient SLAM; implemented voxel hashing for dense mapping and integrated the reconstruction module with robot path planning.
- **Humanoid Robot Control:** Developed bipedal walking patterns on non-planar surfaces by optimizing ZMP trajectories and implemented localized A\* planning for obstacle navigation. ([demo](#))

## HONORS, AWARDS & GRANTS

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- **Research Grant, Toyota Research Institute of North America** 2023 – Present
- **Ira A. Fulton Schools of Engineering Fulton Fellows Award, Arizona State University**
- **Student with Distinction, Arizona State University**

## SKILLS

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- **Programming:** Python, C/C++, CUDA, JavaScript, MATLAB
- **Machine Learning:** PyTorch, deep learning, contrastive learning, representation learning, multi-task learning, generative models, model distillation
- **Computer Vision and Robotics:** Visual odometry, SLAM, VIO, 3D reconstruction, pose estimation, bundle adjustment, sensor fusion, path planning
- **Math and Optimization:** Lie theory, quaternions, nonlinear optimization, geometric optimization, spatial intelligence
- **Tools and Frameworks:** OpenCV, Ceres, Eigen, ROS, Docker, Git, OpenGL, SIMD optimization, CEVA/Hexagon DSP

## LANGUAGES

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- English (Fluent), Mandarin (Native)